

Mental health has been widely recognized in recent years, as the prevalence of mental health disorders continues to rise. In this regard there has been need to come up with screening techniques to ensure accurate and effective screening tools for mental health disorders. PHQ-9 is one the tools that have been employed over time to screen severity of depression. Patient Health Questionnaire (PHQ-9) is a brief, self-administered screening tool that is commonly used to diagnose and assess the severity of depression. The questionnaire consists of nine items that correspond to the nine DSM-5 criteria for major depressive disorder. Each item asks the patient to rate the frequency of a particular symptom over the past two weeks on a scale from 0 (not at all) to 3 (nearly every day). The scores are then summed to give a total score that ranges from 0 to 27, with higher scores indicating greater severity of depression. Levis et al. 2019, evaluates the accuracy of the Patient Health Questionnaire-9 (PHQ-9) for screening major depression in adults using individual participant data meta-analysis. The study, which involved the DEPRESSD Collaboration, analyzed data from 27 studies, with a total of 50,788 participants. Levis et al. found that a cut-off score of 10 or above can be used regardless of age, and the PHQ-9 is similarly sensitive but may be less specific for younger patients than for older patients. The study also found that the PHQ-9 sensitivity compared with semi structured diagnostic interviews was greater than in previous conventional meta-analyses that combined reference standards and suggest that future research on the PHQ-9 should ideally be based on semi structured diagnostic interviews and should consider estimating probabilities of depression across the full spectrum of PHQ-9 screening scores, and should combine screening scores with individual characteristics to generate individualized probabilities of major depression. Levis et al. developed a web-based tool to estimate the expected number of positive screens and true and false screening outcomes based on the study's results. However, in primary care, approximately half of patients with positive screens would be false positives if this was used in practice.